

about how, when and where to move. Machine learning helps them do this quite easily. Apart from this, there are various decisions that robots need to take instantaneously while they perform any activity, which is possible using different machine learning techniques.

6. *Game playing:* A strategy game is one in which the player's autonomous decision-making skills are quite significant in determining the final outcome. Almost all the strategy games require internal 'decision tree style' of thinking, and typically require very high situational awareness. Machine learning meets all these requirements and, hence, is widely used in gaming.

Advantages of machine learning

1. Machine learning techniques help the system to take decisions on the basis of training data in dynamic or uncertain situations as well.
2. It can handle multi-dimensional, multi-variety data, and can even extract implicit relationships within large data sets in a dynamic, complex and chaotic environment.
3. It allows reduction of the time cycle and improves resource utilisation. It also provides different tools for continuous quality improvement in any large or complex process.
4. Another advantage of machine learning techniques is the increased usability of various applications of algorithms due to source programs like Rapidminer. Machine learning allows easy application and the comfortable adjustment of parameters to improve the classification performance.

Challenges of machine learning

1. A very common challenge is the acquisition of relevant data.
2. Once the available data is secured, it often has to be pre-processed depending on the requirements of the specific algorithm used, which has a critical impact on the final results.
3. Sometimes, the interpretation of results also becomes a challenge, as these need to be interpreted according to the algorithm chosen.

Different machine learning techniques can be implemented in order to let the system decide on what action it needs to take and when it needs to be taken. Machine learning can really give an edge to automation, and has already helped in making the world a lot smarter. But machines haven't stopped learning, yet, and the next level of this technology is being worked on as I write this! 

References

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By: Vivek Ratan

The author currently works as an automation test engineer at Infosys, Pune. He can be reached at ratannivek14@gmail.com

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